



Department of Mathematics and Natural Sciences
Mid Term Examination
Fall 2023
Course Title: Elements of Statistics and Probability (STA 201)
Section: 32
Total Marks: 20

Time: 1 hour

Date: November 6, 2023

Special Note: Answer all questions

1. You have data on the annual income (in thousands of dollars) and education level (measured in years of formal education) for 10 individuals:

[4+5+1=10.0]

Person	1	2	3	4	5	6	7	8	9	10
Education Level	12	14	16	10	18	13	15	11	17	12
Income	4*	5*	6*	3*	7*	4*	5*	3*	6*	4*

Here, * = Last digit of your ID

* = Second last digit of your ID

- Construct a scatter plot for the data. Do the data appear to have a positive correlation, a negative correlation, or no correlation? Explain.
- Calculate the Pearson correlation coefficient for the data points. Is there a correlation between variables Education Level and Income? If so, is it a positive or negative correlation?
- Based on the results from questions a and b briefly discuss the relationship of the variables Education Level and Income.

$$r = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sqrt{(\sum_{i=1}^n x_i^2 - n \bar{x}^2)(\sum_{i=1}^n y_i^2 - n \bar{y}^2)}}$$

2. You have a sample of 11 temperatures (°C) recorded in a city over a week: 10, 22, 15, 30, 5, 20, 12, 35, 75, 25, **. Create a box plot and determine if there are any outliers. **[5.0]**

Here, ** = Last two digits of your ID

$[Q_i = (\frac{i-(n+1)}{4})^{th}$ value of the observation, where $i=1,2,3$]

3. You have collected data on the travel times (in minutes) of two different routes to work for a month: **[5.0]**

Route A	18*	20*	22*	19*	21*	23*	18*	22*	20*	21*	22*	19*
Route B	15*	16*	28*	14*	17*	30*	16*	15*	19*	15*	16*	14*

Here, * = Last digit of your ID

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Determine which route has more consistent travel times.

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$